



8-Day Beginner Project Plan

Project: Vacation Budget Calculator

Goal: Track income vs. vacation expenses for 4 people over 7 days



Classroom Requirements Spread Across Project

Requirement	When You'll Practice It
Variables	Day 1–2
Data types (str, int, float, list, dict)	Day 1–4
Loops	Day 3, 5, 7
Decision-making (if/else)	Day 3–8
Lists of dictionaries	Day 4, 5, 6
Functions	Day 3–8
Documentation (comments)	Day 3–8



Day-by-Day Plan



Day 1: Variables + Project Setup

You'll Learn:

- What variables are
- How to name and assign them

You'll Do:

- Create variables for:

- `trip_days = 7`
- `people = 4`
- `budget = 0` (user will update this)
- Start Python file: `vacation_budget.py`
- Print variables using `print()`
- Comment your code to explain what each variable means

✓ *Classroom Skill: Variables, Data Types, Comments*

■ Day 2: More Variables + Simple User Input

You'll Learn:

- `input()` and how to store numbers

You'll Do:

- Ask user to input:
 - `income`
 - `trip_budget`
- Convert string to float: `float(input())`
- Do simple math: `savings = income - trip_budget`
- Print result

✓ *Classroom Skill: Variables, Data Types, Decision-making (if savings > 0)*

Day 3: Add Daily Expenses + If/Else

You'll Learn:

- Simple conditionals (`if` statements)

You'll Do:

- Create 3–5 fixed expenses:
 - `hotel_cost_per_night`, `food_per_day`, `transport`, `tickets`
- Multiply costs by days or people
- Use `if savings >= total_cost:` to print message:
 - + "You can afford the trip!"
 - "You need to save more."

 *Classroom Skill: If/Else logic, calculations, comments*

Day 4: Use Lists + Dictionaries for Expenses

You'll Learn:

- How to store expenses as a list of dictionaries

You'll Do:

- Create:

```
python
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expenses = [
    {"category": "Hotel", "cost": 150},
    {"category": "Food", "cost": 50},
    {"category": "Gas", "cost": 100}
]
```

- Print each item with a `for` loop

✓ *Classroom Skill: Lists of dictionaries, Loops, Variables*

🟡 Day 5: Loop Through Expenses + Calculate Total

You'll Learn:

- `for` loops
- Running totals

You'll Do:

- Use a loop:

```
python
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total = 0
for expense in expenses:
    total += expense["cost"]
```

- Print total and compare it to `budget`

✓ *Classroom Skill: Loops, Decision-making, Variables, Functions (if you wrap in one)*

🟢 Day 6: Save/Load with JSON (Optional if Ready)

You'll Learn:

- Saving and loading data

You'll Do:

- Use `import json`
- Save your `expenses` list to a file
- Load it back and print

✓ *Classroom Skill: Data types, File I/O, Functions*

■ Day 7: Clean Code + Build Menu

You'll Learn:

- Organize code into functions
- Use loops to create a main menu

You'll Do:

- Define:
 - `add_expense()`
 - `calculate_total()`
 - `can_afford_trip()`
- Create `while True:` loop with options:
 - Add expense
 - View total
 - Quit

✓ *Classroom Skill: Functions, Loops, Decision-making*

■ Day 8: Final Testing + Optional Graph

You'll Do:

- Test all inputs and edge cases
- Check for errors (negative amounts, typos)
- Bonus: Add `matplotlib` pie chart for expenses

✓ *Classroom Skill: Debugging, Decision-making, Bonus Visualization*
